

Worksheet solutions and trace key

Instructor copy. Distribute after the discussion, not beside the original conjectures.

1. Main graph

The first discovered target route is $S \rightarrow T$ with cost 15. An optimal route is $S \rightarrow A \rightarrow B \rightarrow C \rightarrow T$, costing $2+1+1+4=8$. Exhaustive enumeration finds 8 simple S-to-T paths. The cycle $A \rightarrow B \rightarrow A$ must be excluded only from the current candidate route, not globally from all candidates.

Non-stale selected node	Improved labels	New predecessor
S, 0	A=2; B=6; D=2; T=15	S for each
A, 2	B=3; C=8	A for B and C
D, 2	E=3	D
B, 3	C=4; T=11	B for C and T
E, 3	F=4	E
C, 4	T=8	C
F, 4	G=5	F
G, 5	None	None
T, 8	Goal extracted; stop	Final path recovered

Lazy Dijkstra pops and skips $B=6$ and $C=8$ before the current $T=8$ entry. $T=11$ and $T=15$ remain unprocessed at early termination. The complete pop sequence is $S_0, A_2, D_2, B_3, E_3, C_4, F_4, G_5, B_6(\text{stale}), C_8(\text{stale}), T_8$.

Proof: Let u be the current minimum non-stale frontier node. On a supposedly cheaper route to u , take its first unfinalized node y and finalized predecessor x . Relaxation from x supplied a frontier label no greater than that optimal prefix cost. The nonnegative suffix means that prefix is no more costly than the whole route to u . This contradicts u 's minimum label if u were suboptimal. The essential inequality is $\delta(y) \leq \delta(u)$. [R1, R4]

Negative-edge refutation: $S \rightarrow T=2$, $S \rightarrow B=5$, $B \rightarrow T=-4$. A Dijkstra-style early goal pop gives 2, but the optimum is 1. The suffix can now reduce cost. The reference API rejects this graph.

2. A* and the reopening counterexample

The main-graph heuristic is consistent. Its A* selections are S($g=0, f=6$), A(2,7), B(3,7), C(4,7), T(8,8). The distracting D branch remains at $f=9$. The terminal target extraction is not counted as an expansion.

For the separate four-node example:

Selected state	g	h	f	Consequence
S	0	0	0	Queue A at $f=3$ and B at $f=5$
A	3	0	3	Discover T at $g=6$
B	1	4	5	Discover improved A at $g=2$
A again	2	0	2	Improve T to $g=5$
T	5	0	5	Return optimum 5

All heuristic values are admissible: $h^*(S)=5$, $h^*(A)=3$, $h^*(B)=4$, $h^*(T)=0$. But the edge $B \rightarrow A$ violates consistency because $4 > 1+0$. Permanent closure suppresses the second A expansion and returns cost 6.

Two repairs: Allow improved routes to reopen expanded states; or establish consistency so the Dijkstra-like finalization argument holds. Under the reference implementation, expanded is for statistics only.

Reweighting: $S \rightarrow A$ becomes 3; $S \rightarrow B$ becomes 5; $B \rightarrow A$ becomes -3; $A \rightarrow T$ remains 3. The negative reweighted edge explains the same finalization failure. This does not make the original edge weights negative.

Overestimation test on the main graph: Changing $h(A)$ to 20 hides the optimal prefix. The reference A* returns cost 11 via $S \rightarrow B \rightarrow C \rightarrow T$. $h^*(A)=6$, so $h(A)=20$ violates admissibility.

3. Movement model and exit ticket

For start (0,0) and target (1,1), a unit-cost diagonal costs 1 while Manhattan gives 2. On an unobstructed eight-neighbor grid with all moves costing 1, Chebyshev distance $\max(|dx|, |dy|)$ is the exact relaxed cost. A different diagonal price requires a different argument; do not reuse this formula without the move-cost assumption.

Exit A: Goal discovery gives the cost of a known route, an upper bound on the optimum. Correct non-stale goal extraction is certified only under the algorithm's assumptions: nonnegative costs for Dijkstra; admissibility plus correct improvement propagation for reopening A*, or a suitable consistent-heuristic finalization implementation.

Exit B: A stale record was superseded by a better route already known. Reopening propagates a new, genuinely improved route after an earlier expansion. Discarding the former and allowing the latter are compatible and necessary in the lazy reopening A* implementation.

Exit C: Expansion count ignores heuristic-evaluation cost, queue operations, implementation constants, and per-state neighbor-generation cost. The supplied benchmarks also include initialization and path recovery. Runtime must be measured separately from animation speed.

References: docs/references.md. Exact conventions: docs/technical_notes.md.